Discrete Mathematics and Statistics (CPT107)

Tutorial 4

1: Show that the relation "is a divisor of" on the set $\mathbb{Z}^+$ of positive integers is a partial order.

2: Suppose that A, B, E, F, G, H, I and K represent people on twitter. We assume that everybody follows himself on twitter. Also, everybody follows G. Everybody follows E except H, I, F and G. Everybody follows F except E and G. Also, A follows B and K follows everybody. It turns out that the relation "follows" is a partial order in this case. Draw the Hasse Diagram of this partial order. Construct a pair of people, such that, if the first of them started following the second, then "follows" would no longer be a partial order.

3: Let $M = \{a, b, c\}$. What is the transitive closure of the relation $\{(a, a), (a, b), (a, c), (c, a), (b, c)\}$ on M?

4: Define the relation on ℝ x ℝ by (a,b) R (x,y) iff a ≤ x and b ≤ y. Prove that R is a partial order on ℝ x ℝ.

5: Given a partially ordered set, is a pair $\mathbf{P} = (X, \leq)$, where X is a nonempty set and $\leq$ is a partial order on X; that is, for x, y, and $z \in X$
(1) $x \leq x$ (reflexivity)
(2) $x \leq y$ and $y \leq x$ imply $x = y$ (antisymmetry)
(3) $x \leq y$ and $y \leq z$ imply $x \leq z$ (transitivity)

Prove that: If $(X, \leq)$ is a finite partially ordered set, then X has a maximal and a minimal element.

6: Prove that the divides relation on $A = \{1,2,4,8,\dots,2^n\}$ is a total order; where n is a nonnegative integer.